Exp No: 6 – Time series analysis of website traffic

Program:

```
import pandas as pd

import matplotlib.pyplot as plt

from statsmodels.tsa.seasonal import seasonal_decompose

# Load dataset

data = pd.read_csv('website_traffic.csv', parse_dates=['Date'], index_col='Date')

# Time Series Decomposition

decomposition = seasonal_decompose(data['Page_Views'], model='additive')

decomposition.plot()

plt.show()

# Moving Average

data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()

data[['Page_Views', 'Page_Views_MA']].plot()

plt.title('Page Views with Moving Average')

plt.show()

# Anomaly Detection

data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) /
data['Page_Views'].std()

anomalies = data[data['Page_Views_Z'].abs() > 3]

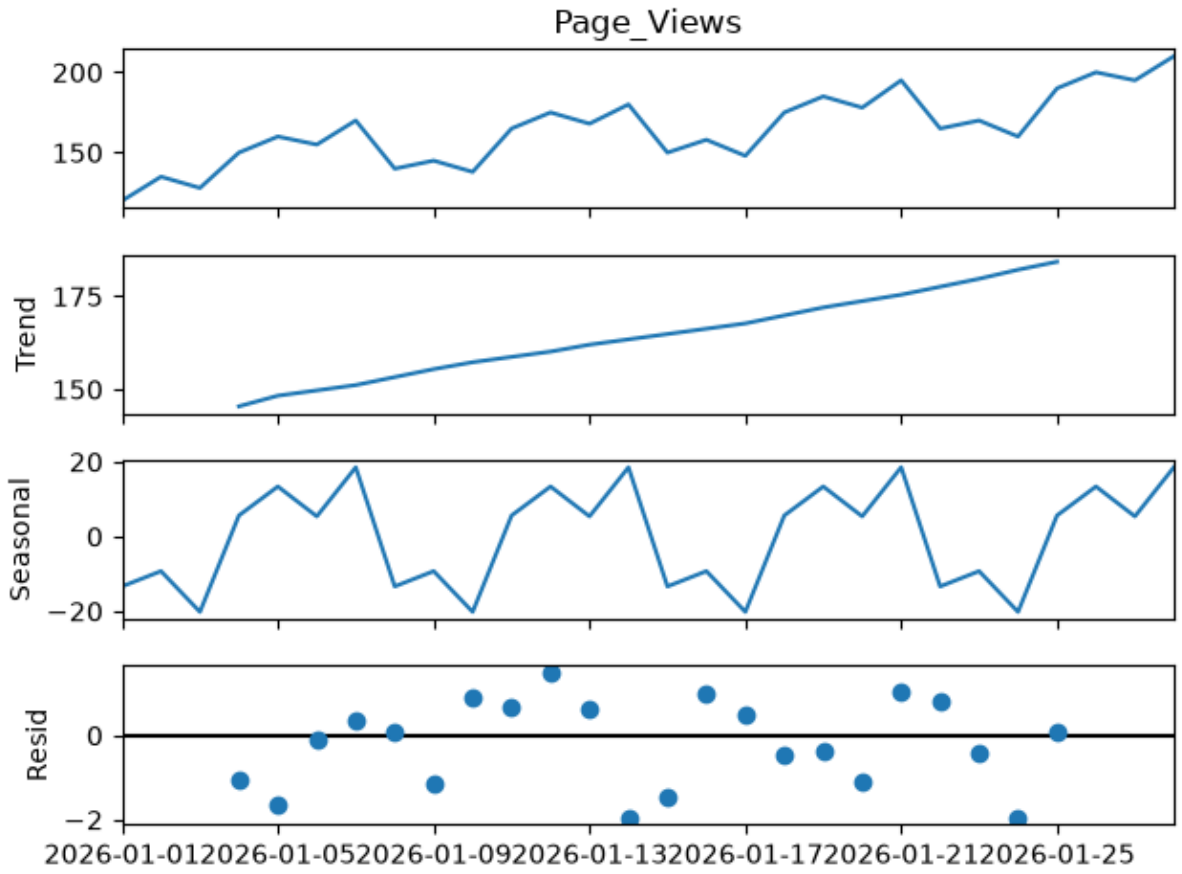
plt.plot(data.index, data['Page_Views'], label='Page Views')

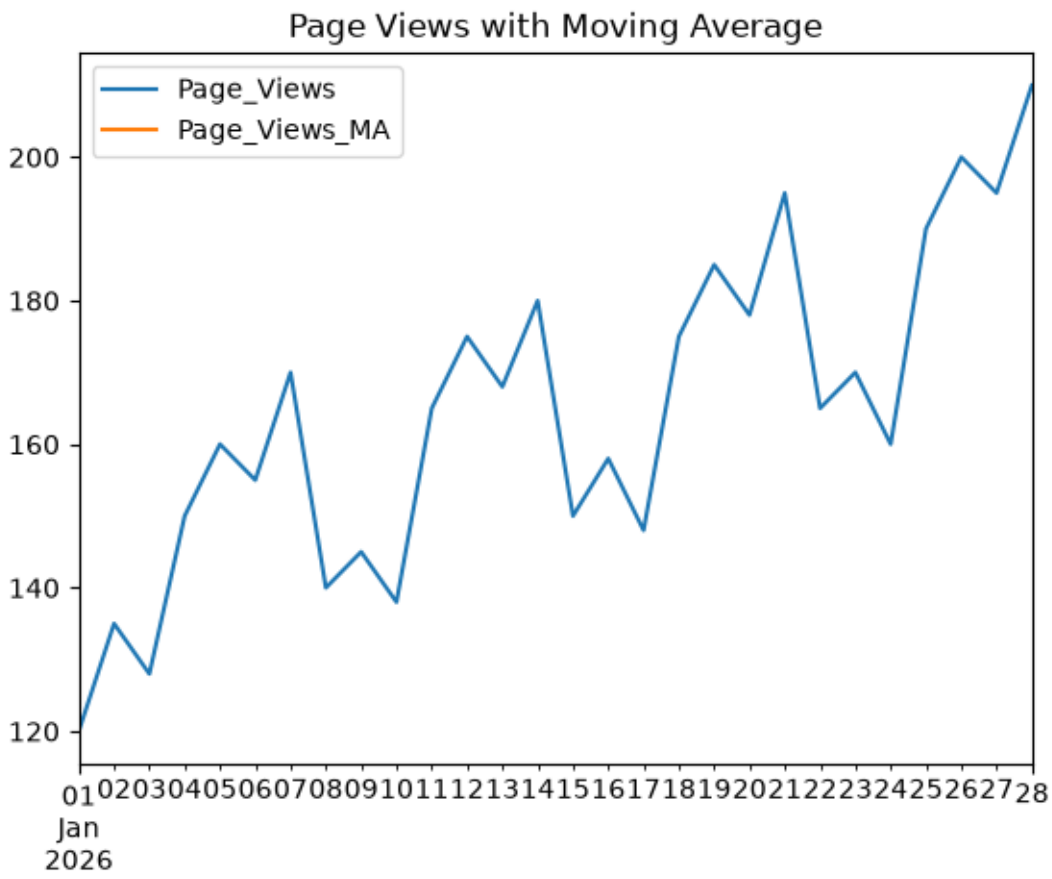
plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomalies')

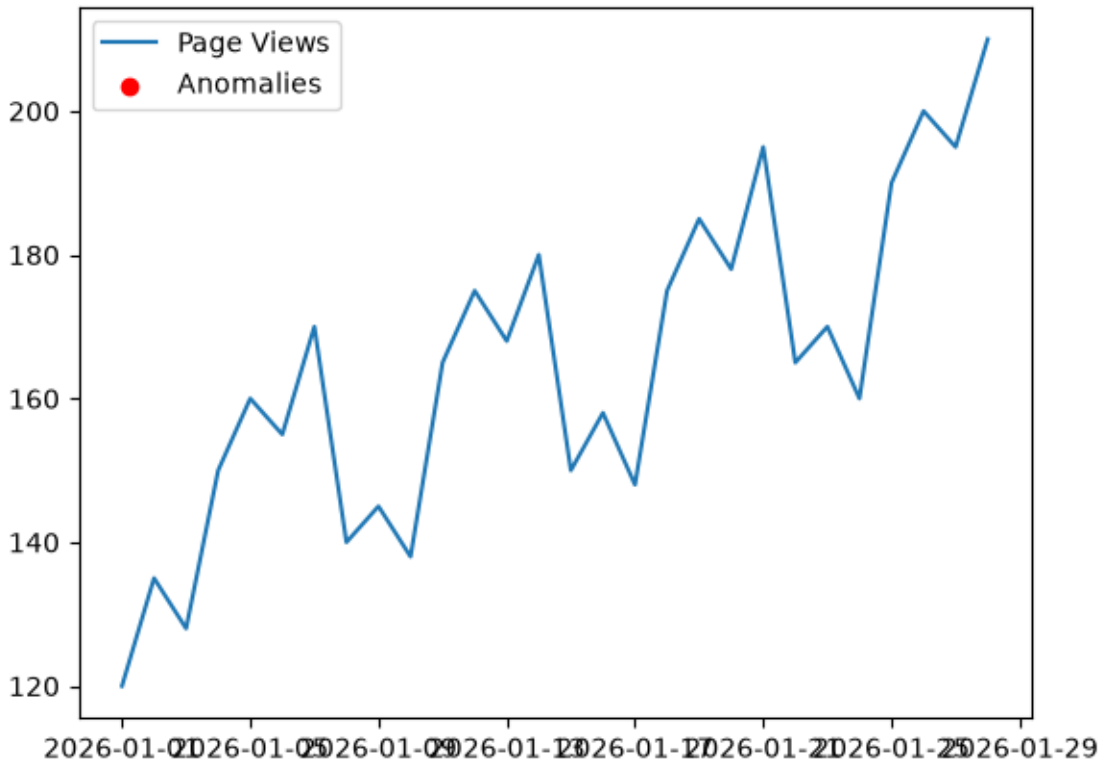
plt.legend()

plt.show()
```

Output:


Page_Views
200
150
Trend
175
150
Seasonal
20
0
−20
Resid
0
−2
2026-01-01
2026-01-05
2026-01-09
2026-01-13
2026-01-17
2026-01-21
2026-01-25


Page Views with Moving Average
Page_Views
Page_Views_MA
200
180
160
140
120
01
Jan
2026
02030405060708091011121314151617181920212223242526272 8
Date


Page Views
Anomalies
200
180
160
140
120
2026-01-21
2026-01-25
2026-01-29